The Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

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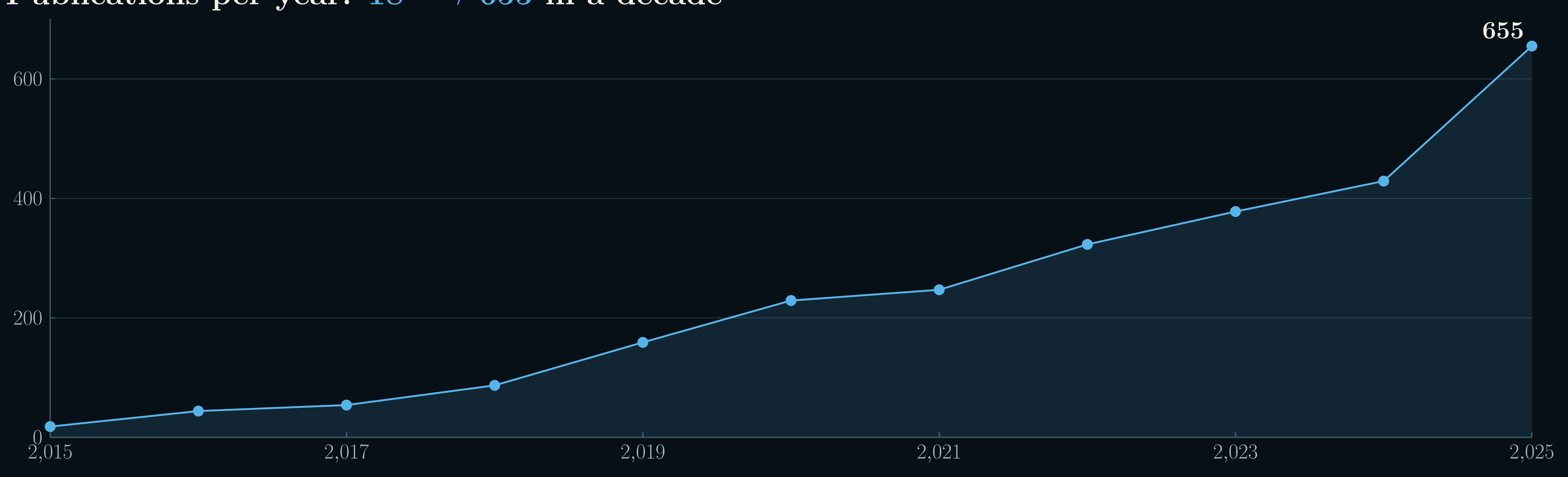
Perturbation evidence is growing faster than it can be organised

Publications per year: 18 ⟶ 655 in a decade



Europe PMC yearly publication counts for title/abstract mentions of CRISPR screens, Perturb-Seq, deep mutational scanning or multiplexed assays of variant effect; snapshot 24 August 2026.

36×
more papers per year than in 2015.

>2.5 million
cells profiled in a single genome-scale Perturb-Seq study. Replogle et al., *Cell* 2022; doi:10.1016/j.cell.2022.05.013.

This evidence is exceptionally rich, but studies arrive with different repositories, identifiers, metadata depth, score semantics, gene annotations and processing choices. The result is abundance without straightforward reuse.

The Catalogue turns scattered studies into one coherent resource

What arrives

Papers, supplements, repository records, matrices and raw sequencing files

Author-specific formats and processing decisions

⟶

What the Catalogue standardises

One validated metadata schema; Ensembl gene identity; ontology-aligned context; modality-appropriate analysis; explicit provenance

BigQuery and dbt harmonise analytical tables; Elastic and PostgreSQL serve search and results

⟶

What users can retrieve

Search by target or dataset; REST API; JSON metadata; complete CSV and Parquet files; interactive modality views

The same identifiers, metadata contract and access methods across modalities

The schema preserves real biological differences—model system, assay, perturbation mechanism, time point and readout—while removing avoidable technical friction. For selected Perturb-Seq studies, raw FASTQ is reprocessed through a shared workflow covering counts, guide assignment, quality control, differential expression and Hallmark gene-set enrichment.

Today, 1,237 datasets span three complementary modalities

1,237 datasets
19,235 human targets
36 tissues
34 cell types
1,200 cell lines
169 diseases

CRISPR screens — 1,201 datasets

Gene-level phenotype and fitness scores, harmonised significant hits and rich experimental context.

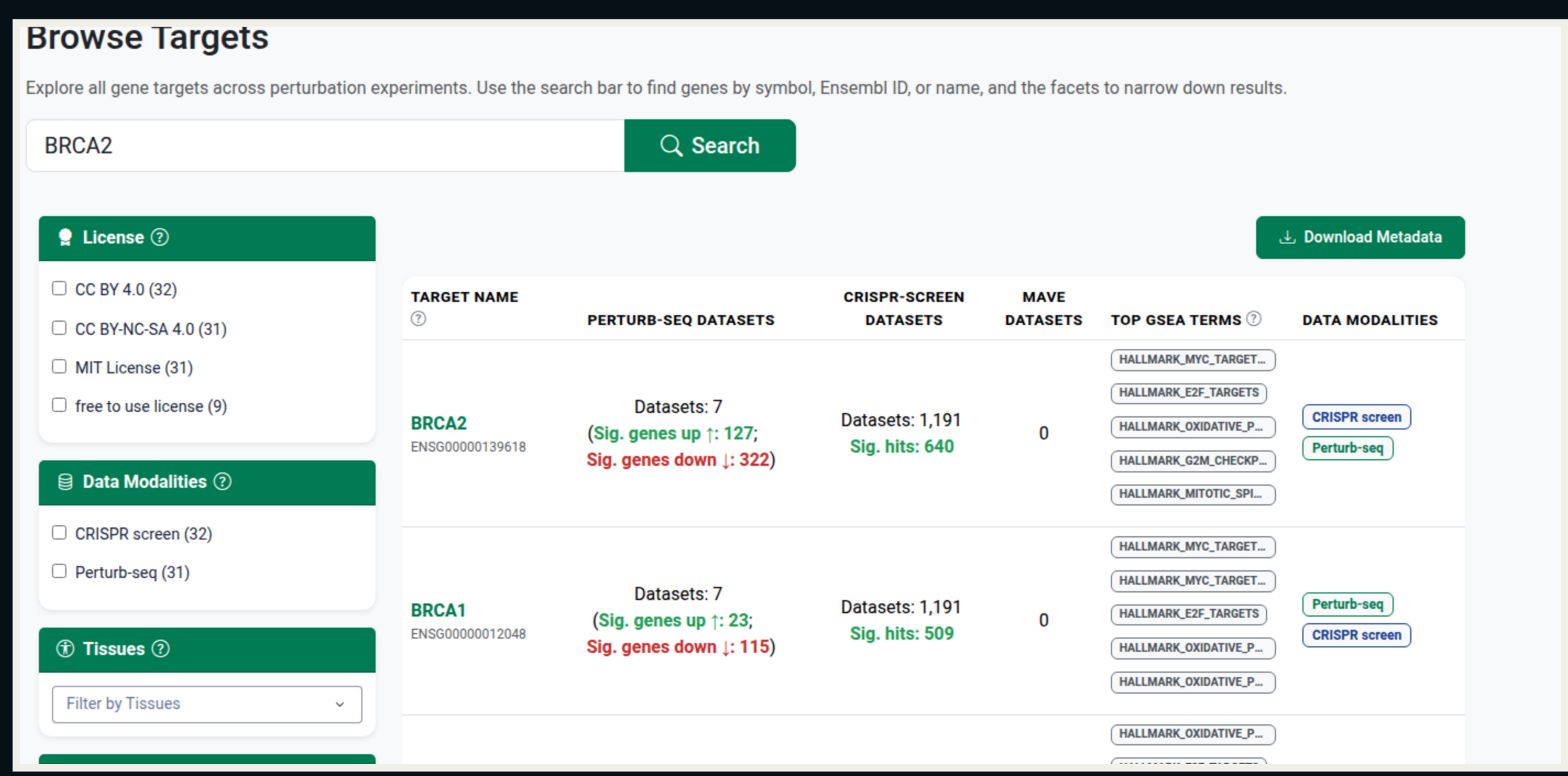
Perturb-Seq — 26 datasets

Single-cell transcriptomic responses, including differential expression and pathway enrichment. Five datasets have been reprocessed from raw FASTQ through the shared pipeline.

MAVE — 10 datasets

Variant-level functional measurements across protein sequence, with searchable scores and effect heatmaps.

Production snapshot: 24 August 2026. All modalities use normalised Ensembl gene IDs.



The live BRCA2 target view brings modality coverage, contributing datasets, significant effects and enriched pathways together.

Next, we will make integration continuous and reanalysis broader

Curate faster, keep human judgement

AI-assisted extraction will draft structured metadata from studies and repositories. Curators will review biological interpretation, resolve ambiguity and validate ontology mappings before release.

Reanalyse more, measure quality

We will extend kit and experimental-design support, develop author-independent quality metrics and broaden modality-appropriate reanalysis beyond the first five Perturb-Seq datasets.

Release predictably, preserve provenance

Periodic, versioned releases will make additions and analytical changes explicit, reproducible and easier to adopt in downstream research and production workflows.

Help shape what comes next

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations that prioritise modalities, reanalysis, datasets or functionality relevant to your programme.
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Explore the Catalogue

EMBL-EBI

Open Targets